Spring 2026

NE 523 – Computational Transport Theory (3 CH)

Prof. Yousry Y. Azmy
2146 BU, (919) 515-3385, yyazmy@ncsu.edu
Schedule: MW 10:15-11:30 am, Room 327 at 111 Lampe Drive

Prerequisites

- NE 401/501: Reactor Analysis and Design
- Advanced math & moderate programming skills are necessary. Permissible programming languages: Fortran or C++

Student Learning Outcomes

- Explain the physics foundation of the neutron transport equation (NTE) and its underlying assumptions
- Identify relationships among various numerical methods employed in solving the NTE
- Analyze and justify discretization methods in energy, angle, and space for computationally solving the NTE
- Apply spectral analysis to predict performance of iterative algorithms for solving the NTE
- Implement new Fortran or C++ code or maintain existing code for solving the NTE

Textbook

- E. E. Lewis and W. F. Miller, *Computational Methods of Neutron Transport*, American Nuclear Society (1993) - $70

Course Overview

Derivation of the nonlinear Boltzmann equation for a rarefied gas and linearization to the equation of transport of neutral particles. Deterministic methods for solving the neutron transport equation: Multigroup energy discretization; Discrete Ordinates angular discretization; various spatial discretization methods. Convergence of numerical solutions with discretization refinement. Iterative solution algorithms: inner, outer, and power iterations. Spectral analysis of inner iterations convergence and acceleration. Selection of advanced topics.

Course Structure

Two lectures per week

Grade Distribution

Homework Assignments	20%
Code Assignments	20%
Quizzes	20%
Midterm	20%
Final Exam	20%

Policy on late submission of Homework & Code Assignments:

- Up to one week past due date ⇨ Grade reduced by 50%
- No credit for homework submitted more than one week past the due date

Attendance

Required but not enforced or tracked

Reference Material

- G. I. Bell and S. Glasstone, *Nuclear Reactor Theory*, Van Nostrand Reinhold Co. (1970)
- J. J. Duderstadt and L. J. Hamilton, *Nuclear Reactor Analysis*, Wiley (1976)
- H. Greenspan, C. N. Kelber, and D. Okrent (eds.), *Computational Methods in Reactor Physics*, Gordon and Breach Science Publishers (1968)
- Kerson Huang, *Statistical Mechanics*, John Wiley & Sons, Inc. (1963).
- John W. Dettman, *Mathematical Methods in Physics and Engineering*, McGraw-Hill Book Company (1969).

Statement for Students with Disabilities

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with Disability Services for Students at 1900 Student Health Center, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation (REG 02.20.01).

N.C. State University Polices, Regulations, and Rules (PRR)

Students are responsible for reviewing the PRRs which pertain to their course rights and responsibilities. These include: http://policies.ncsu.edu/policy/pol-04-25-05 (Equal Opportunity and Non-Discrimination Policy Statement), http://oied.ncsu.edu/oied/policies.php (Office for Institutional Equity and Diversity), http://policies.ncsu.edu/policy/pol-11-35-01 (Code of Student Conduct), and http://policies.ncsu.edu/regulation/reg-02-50-03 (Grades and Grade Point Average).

Artificial Intelligence

Artificial Intelligence will be an important tool in your life, but it is not yet ready to replace the engineering thought process that this class emphasizes. It is an academic integrity violation to use AI without proper authorization and attribution. This course allows a strictly restrictive use of artificial intelligence (AI) tools, such as chatbots and text generators to get guidance on coding assignments/homeworks, as long as you do so in an ethical and responsible manner. Essentially, you can think of these tools as ways to help you learn/find something related to coding practices instead of using search engines and other coding forums. This means that:

- You **must not** use AI tools to replace your own thinking or analysis or to avoid engaging with the course content.
- You must **cite** or explain any AI tools you use. Provide the name of the AI tool, the specific prompt or query you used to generate the output.
- You must be **transparent** and **honest** about how you used the AI tool and how it contributed to your solution of the assignment.
- You must **explain** what you learned from the AI tool, how you verified its accuracy and reliability, how you integrated its output with your own work, and how you acknowledged its limitations and biases.
- You will be held **accountable** for any mistakes or errors made by the AI tool.

Using AI tools in an unethical or irresponsible manner, such as copying or paraphrasing the output without citation or transparency, using the output as your own work without verification or integration, or using the output to misrepresent your knowledge or skills, is considered a form of academic dishonesty and will result in a zero grade for the assignment and possible disciplinary action. If you have any questions about what constitutes ethical and responsible use of AI tools, please consult with the instructor before submitting your work.

NE 523 Syllabus – Course Schedule

Spring 2026†

#	Date	Subject	HW/Code	Exam
1	1/12	Course Intro/preliminaries; Scope of Boltzmann Eq.		
2	1/14	Derivation of Boltzmann Equation		
	1/19	**MLK Holiday – No Classes**		
3	1/21	Collision term & *Molecular Chaos*		
4	1/26	Multispecies Nonlinear Boltzmann Eq. (NBE)	Assign HW 1	
5	1/28	Neutron Transport Eq (TE) as linearized Boltzmann Eq.	Assign Code 1	
6	2/2	Basic definitions; TE Assumptions	HW 1 Due	
7	2/4	Initial & Boundary Conditions; Vector Spaces	Code 1 Due	
8	2/9	Scalar Products		
9	2/11	Orthonormal Basis; Scat$^{\underline{g}}$ Source: Spherical Harmonics	Assign HW 2	Quiz 1
10	2/16	TE numerical meth; Energy discretization & MG *const*$^{\underline{s}}$		
11	2/18	Angle Discretization: S_N in slab, multi-dimensional	HW 2 Due	
12	2/23	S_N spatial discretization: Multi-D DD		
13	2/25	Iterations: Multi-D mesh sweep	Assign Code 2	Quiz 2
14	3/2	Iterations: Inner; Integral Form of TE		
15	3/4	Integral TE: Isotropic & anisotropic scattering	Code 2 Due	Midterm
16	3/9	Criticality: SS solution; k-ev; k- *vs* α-eigenproblem		
17	3/11	Iterations: Outer; Power	Assign Code 3	MT Due
3/16 – 20		**Spring Break – No Classes**		
18	3/23	Adjoint basics; Adjoint non-multiplying TE; Importance	Assign HW 3	
19	3/25	Existence & uniqueness of TE solution	Code 3 Due	Quiz 3
20	3/30	Invariance & symmetry: Dimension & size reduction	HW 3 Due	
21	4/1	Asymptotics; Numerical solution convergence	Assign Code 4	
22	4/6	Convergence of TE numerical solutions: E, $\mathbf{\Omega}$, $\mathbf{r}$		
23	4/8	S_N solution: existence, regularity	Code 4 Due	Quiz 4
24	4/13	S_N solution: uniqueness, non-negativity, convergence	Assign HW 4	
25	4/15	Error analysis of Diamond Difference method		
26	4/20	Spectral analysis of iterations; Eigenmode convergence	HW 4 Due	Final
27	4/22	Spectral analysis of inners & DSA; Step spatial discret$^{\underline{n}}$		
28	4/27	Spatial discretization: CC, Nodal, FE – DGFEM/LD	Final Due	Quiz 5
Extra Class		Ray Effects in S_N; Splicing & Bootstrapping; Neg$^{\underline{ve}}$ fixup		

† Course schedule is subject to change with appropriate notification to students